THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Seeds dispersed by wind have: 1

(a) no coat **(b)** fleshy covering **(c)** hooks **(d)** wings or hairs

2. The troposphere is warmer near the ground mainly because: 1

(a) ozone absorbs UV
(b) the ground is heated by sunlight and warms the air
(c) greenhouse gases are absent
(d) the Sun is closer

3. A metre scale has a worn end and reads 0.3 cm at the start. A measured length reads 5.7 cm. The corrected length is: 1

(a) 5.7 cm **(b)** 0.3 cm **(c)** 6.0 cm **(d)** 5.4 cm

4. The solubility of KNO3 is 64 g/100 g water at 40 degrees Celsius and 32 g at 20 degrees Celsius. 164 g of saturated solution at 40 degrees Celsius is cooled to 20 degrees Celsius. The mass of crystals is: 1

(a) 32 g **(b)** 16 g **(c)** 96 g **(d)** 64 g

5. A sample of a compound contains 2.4 g of carbon and 6.4 g of oxygen. The ratio C : O by mass and the likely formula (C = 12, O = 16) are: 1

(a) 3 : 4, CO **(b)** 3 : 8, CO_2 **(c)** 3 : 16, CO_3 **(d)** 1 : 2, CO_2

6. Two ions X^{2+} and Y^{2-} each have 18 electrons. The atomic numbers of X and Y are respectively: 1

(a) 18 and 18 **(b)** 19 and 17 **(c)** 16 and 20 **(d)** 20 and 16

7. A car goes from rest with acceleration a for time t, then retards uniformly with 2a till it stops. The total time is: 1

(a) 2 t **(b)** 1.5 t **(c)** 3 t **(d)** t

8. Assertion (A): Action and reaction forces never cancel each other. Reason (R): They act on different bodies. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): The kinetic energy of a body becomes four times when its speed is doubled. Reason (R): KE is proportional to the square of the speed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The speed of sound is greater in solids than in gases. Reason (R): The particles in solids are closely packed and strongly bound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Plant cells do not burst when placed in distilled water. Reason (R): The cell wall resists excessive expansion of the cell. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. A plant tissue is stained with safranin and some cells turn red. They contain: 1

(a) lignin **(b)** starch **(c)** protein **(d)** cellulose only

13. Dolphins and sharks look similar but belong to different classes because: 1

(a) dolphins have lungs, hair and mammary glands

(b) sharks have lungs

(c) dolphins have gills

(d) both have fins

14. The female reproductive structure that is the site of fertilisation and passage of the egg is the: 1

(a) cervix **(b)** oviduct **(c)** uterus **(d)** vagina

15. Assertion (A): Melting of polar ice accelerates global warming. Reason (R): Water has a lower albedo than ice. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

16. Which situation requires an exact value rather than an estimate? 1

(a) height of a mountain in rough terms **(b)** number of stars in a galaxy

(c) population of a country **(d)** dosage of a medicine for a child

17. Which of the following combinations can be separated by magnetic separation? 1

(a) water and oil **(b)** iron and sulphur **(c)** salt and sugar **(d)** sand and chalk

18. An element X forms a chloride XCl_2 with 52.5 per cent X by mass? (Cl = 35.5) The atomic mass of X is closest to: 1

(a) 56 **(b)** 40 **(c)** 24 **(d)** 78.5

19. A sample of boron (average mass 10.8 u) has isotopes 10 and 11. The percentage of the lighter isotope is: 1

(a) 10.8 **(b)** 50 **(c)** 80 **(d)** 20

20. A man walks to a market at 4 km/h and returns at 6 km/h. The average speed is: 1

(a) 4.8 km/h **(b)** 5.5 km/h **(c)** 5 km/h **(d)** 4.5 km/h

21. A force of 12 N acts on a body of mass 3 kg for 5 s starting from rest. The final speed is: 1

(a) 60 m/s **(b)** 4 m/s **(c)** 20 m/s **(d)** 12 m/s

22. A body falls from rest from a height h. At what height is its speed half of the final speed? 1

(a) h/2 **(b)** h/4 **(c)** 3h/4 **(d)** h/8

23. A person standing at 340 m from a cliff fires a gun. The echo is heard after: 1

(a) 4 s **(b)** 1 s **(c)** 2 s **(d)** 0.5 s

24. Two cells A and B are placed in the same solution. A shrinks and B swells. This shows that the solution is: 1

(a) hypertonic to A and hypotonic to B **(b)** isotonic to both

(c) hypotonic to both **(d)** hypertonic to both

25. Assertion (A): The trunk of a tree becomes thicker every year. Reason (R): Cambium divides and adds secondary xylem and phloem. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

26. Assertion (A): Bryophytes are called the amphibians of the plant kingdom. Reason (R): They need water for the fertilisation. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

27. Assertion (A): Cross-pollination leads to variation. Reason (R): It involves the fusion of gametes from different plants. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

28. If the albedo of the Earth decreases, the Earth's temperature will: 1

(a) decrease **(b)** increase **(c)** become zero **(d)** remain the same

29. Assertion (A): Estimation is useful in science. Reason (R): It gives a quick check on whether an answer is reasonable. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

30. Assertion (A): Fractional distillation is used to separate the components of air. Reason (R): The gases in liquid air have different boiling points. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

31. Assertion (A): Sodium chloride conducts electricity in the molten state but not in the solid state. Reason (R): In the molten state, the ions are free to move. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

32. Assertion (A): Chlorine has an atomic mass of 35.5 u. Reason (R): Chlorine is made of two isotopes with whole-number masses 35 u and 37 u in a 3 : 1 ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

33. Assertion (A): The average velocity of a body in a round trip is zero. Reason (R): The net displacement is zero. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

34. A heavy box is pushed along the floor with a force F. Which of the following would make the push easier? 1

(a) increasing the weight
(b) reducing the friction by rollers
(c) pushing downward
(d) using a rough floor

35. A bob of a simple pendulum is displaced and released from 0.8 m height above the lowest point. The speed at the lowest point is: 1

(a) 4 m/s **(b)** 2 m/s **(c)** 16 m/s **(d)** 8 m/s

36. A sound of 20 degrees Celsius air moves at 344 m/s. The distance travelled in 0.25 s is: 1

(a) 1376 m **(b)** 43 m **(c)** 86 m **(d)** 172 m

37. A student finds a cell with no nucleus, a cell wall and DNA free in the cytoplasm. The cell is: 1

(a) fungal **(b)** bacterial **(c)** animal **(d)** plant

38. After removing a ring of bark, the leaves above remain fresh for some time because: 1

(a) cork is intact
(b) xylem is intact
(c) phloem is intact
(d) cambium is intact

39. A student finds a plant with a vascular system, seeds in cones and no flowers. The class is: 1

(a) Pteridophytes **(b)** Gymnosperms **(c)** Bryophytes **(d)** Angiosperms

40. The egg released on 14 March remains fertilisable for about 1 to 2 days. The most fertile days are: 1

(a) 1 to 5 March **(b)** 28 to 30 March **(c)** 20 to 25 March **(d)** 13 to 15 March

41. **Assertion (A):** The species is the basic unit of classification. 1
Reason (R): Members of a species can interbreed to produce fertile offspring.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Offspring produced by asexual reproduction are genetically identical to the parent. 1
Reason (R): Asexual reproduction involves a single parent and no fusion of gametes.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** The albedo of fresh snow is high. 1
Reason (R): Snow reflects most of the incoming solar radiation back into space.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The least count of a metre scale marked in millimetres is 1 mm. 1
Reason (R): The metre is the SI unit of length.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Fractional distillation is used to separate two miscible liquids whose boiling points differ by less than 25 K. 1
Reason (R): Simple distillation can separate a volatile liquid from a non-volatile dissolved solid.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** In water, hydrogen and oxygen are always present in the fixed mass ratio 1 : 8. 1
Reason (R): Water is a compound of hydrogen and oxygen.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Rutherford's nuclear model could not explain the stability of the atom. 1
Reason (R): The neutron was discovered by James Chadwick in 1932.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The slope of a position-time graph gives the velocity of the body. 1
Reason (R): The area under a velocity-time graph gives the displacement of the body.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The SI unit of force is the newton. 1
Reason (R): Momentum is the product of the mass and the velocity of a body.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The joule is the SI unit of energy. 1
Reason (R): Power is the rate of doing work.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**